



Multimetallic Rh-La/M/SiO₂ (M = W, V, Ce, and Zr) catalysts for oxygenates synthesis from syngas

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HIGHLIGHTS

- A series of co-promoted Rh-based catalysts were studied for syngas to oxygenates.
- All the catalysts exhibited remarkable selectivity towards oxygenates at ≤ 280 °C.
- Rh has transformed to Rh⁺ and Rh⁰ species from Rh³⁺ state during the reaction.
- The surface Rh⁺ species mainly contribute to the conversion of syngas to oxygenates.

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ABSTRACT

In this study, we investigated a series of co-promoted Rh-La/M/SiO₂ (RLMS, where M = W, V, Ce, and Zr) catalysts for the synthesis of oxygenates from syngas. It was found that all the catalysts showed an excellent selectivity towards oxygenates by suppressing the hydrocarbons formation at lower reaction temperatures (≤ 280 °C). Remarkably, the Rh-La/V/SiO₂ catalyst exhibited the highest oxygenates selectivity (76.5%) among the investigated catalysts at 260 °C. However, at 280 °C, the W-, Ce-, and Zr-promoted Rh-La/SiO₂ catalysts showed better oxygenates selectivity than the Rh-La/V/SiO₂. Rhodium existed as Rh₂O₃ on the surface of the fresh catalysts, which is transformed to Rh⁺ and Rh⁰ species during the reaction. The strong synergistic effects of Rh with the promoters on the surface of the catalysts, which hindered the complete reduction of Rh³⁺ to metallic Rh species, could be an important reason for the higher selectivity towards oxygenates.

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1. Introduction

Catalytic conversion of synthesis gas (syngas) to oxygenates has attracted significant attention in recent years for the effective utilization of coal, natural gas, and biomass towards the production of high-value chemicals and fuels (Zhang et al., 2019). Although several catalytic systems have been investigated, poor oxygenates selectivity found to be a major obstacle for this process (Luk et al., 2017; Ao et al., 2018). Hence, the development of highly active catalysts with satisfactory oxygenates selectivity remains a major challenge.

Supported Rh-based catalysts have been reported as good candidates to date for the selective production of oxygenates via syngas conversion due to their ability to catalyze both CO dissociation and CO insertion steps (Ao et al., 2018; Yang et al., 2016). Despite these promising results, the selectivity to oxygenates over

monometallic Rh-based catalysts is constrained by the thermodynamically favored methane production (Yang et al., 2016; Choi and Liu, 2009). Thus, the suppression of undesired hydrocarbon production is highly desirable to maximize oxygenates selectivity. Extensive studies have revealed that the oxygenates selectivity can be improved by the addition of promoters to supported Rh-catalysts (Borer and Prins, 1993; Subramani and Gangwal, 2008; Subramanian et al., 2010). Particularly, lanthanum has been shown to be an effective promoter for Rh/SiO₂ to enhance the oxygenates selectivity (Borer and Prins, 1993). Studies on the effect of promoters have advanced to multimetallic systems which showed further enhancement of the selectivity towards oxygenates by inhibiting the hydrocarbon selectivity. For instance, the doubly promoted La-V-Rh/SiO₂ catalyst has been reported to exhibit higher oxygenates selectivity in comparison to its singly promoted counterparts (Subramanian et al., 2010). Hence, it can be speculated that the addition of a second promoter together with La to the Rh/SiO₂ based catalyst could be beneficial to increase syngas conversion towards oxygenates formation.

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In this work, we report a series of La/M (M = W, V, Ce, and Zr) co-promoted multimetallic Rh/SiO₂ catalysts, aiming at improving the catalytic performance for the production of oxygenates by syngas conversion. We found that La and all other metals showed a significant promotional effect in enhancing the selectivity towards oxygenates at lower reaction temperatures (≤ 280 °C). The Rh³⁺ state of fresh catalysts didn't undergo complete reduction to metallic Rh during the reaction, which was beneficial for better oxygenates selectivity of the catalysts.

2. Experimental section

2.1. Preparation of catalysts

A series of co-promoted Rh/SiO₂ catalysts (Rh-La/M/SiO₂, where M = W, V, Ce, and Zr) catalysts were prepared by adopting a co-sequential incipient wetness technique. SiO₂ (Alfa Aesar) was used as support material to prepare these catalysts. Rhodium nitrate (Rh (H₂O)(OH)_{3-y}(NO₃)_y, y = 2–3, ~36%, Aldrich), lanthanum nitrate (La (NO₃)₃·6H₂O, Aldrich), ammonium paratungstate hydrate (H₄₂N₁₀-O₄₂W₁₂·xH₂O, 99.99%, Aldrich), ammonium metavanadate (H₄NO₃V, Aldrich), cerium nitrate hexahydrate (Ce(NO₃)₃·6H₂O, Aldrich), and zirconyl nitrate hydrate (ZrO(NO₃)₂·xH₂O, Aldrich), were used as the source of rhodium, lanthanum, tungsten, vanadia, ceria, and zirconia, respectively. The numbers in the parenthesis indicate the Rh/La atomic ratio and the weight percentage of the third metal relative to the initial weight of the support. The third metal M (M = W, V, Ce, and Zr) loading was selected as 1.8 wt% based on the weight of SiO₂. The Rh and La loadings were selected as 1.5 wt% and 2.5 wt%, respectively, in all the formulations, based on the weight of 1.8 wt.%M/SiO₂. For convenience, we have given as Rh-La(x)/y(M)/SiO₂, where x indicates the atomic ratio of rhodium/lanthanum, y indicates the weight percentage of the third metal relative to the initial weight of the support. In a typical synthesis, the required quantity of the third metal (M) precursor was added to a beaker containing SiO₂ support in 50 ml deionized water, followed by drying at 70 °C with continuous stirring until complete evaporation of water and then oven-dried at 120 °C for 12 h. The oven-dried M/SiO₂ was further impregnated with mixed aqueous solution containing Rh and La precursors, followed by oven-dried at 120 °C for 12 h. The resulting materials were ground to obtain homogeneous powder and calcined in a tubular oven at 450 °C for 4 h under continuous air flow (150 ml min⁻¹). The final obtained Rh-La(0.8)/1.8W/SiO₂, Rh-La(0.8)/1.8V/SiO₂, Rh-La(0.8)/1.8Ce/SiO₂ and Rh-La(0.8)/1.8Zr/SiO₂ catalysts were denoted as RLWS, RLVS, RLCS, RLZS, respectively.

2.2. Characterizations

The BET surface areas were obtained by N₂ adsorption on a Micromeritics ASAP 2010 Instrument. Prior to the analysis, samples were oven dried at 120 °C for 12 h and flushed with Argon for 2 h. The pore size distribution analyses were conducted by N₂ physisorption at liquid N₂ temperature using Micromeritics ASAP 2010 apparatus. All samples were degassed at 300 °C under vacuum before analysis.

H₂-temperature-programmed reduction (H₂-TPR) of various catalyst samples were performed by means of an automated catalyst characterization system (Micromeritics, model AutoChem II 2910), which incorporates a thermal conductivity detector (TCD). The experiments were carried out at a heating rate of 5 °C/min. The reactive gas composition was H₂ (10 vol%) in Helium. The flow rate was fixed at 10 ml/min (STP). The sample was then held at 50 °C under flowing helium to remove the remaining adsorbed oxygen until the TCD signal returned to the baseline. Subsequently,

the TPR experiments were performed up to a temperature of 800 °C. The water formed during the reduction was removed by using an ice trapper. The gas stream coming from the reactor was passed through a trap before the gas entered the GC. A mixture of isopropanol and liquid nitrogen was used in the trap to remove the formed water during the TPR experiment.

The X-ray photoelectron spectroscopy (XPS) measurements were performed on a Pyris VG Thermo Scientific spectrometer using Al K α (1486.6 eV) radiation as the excitation radiation. Charging of the catalyst samples was corrected by setting the binding energy of the adventitious carbon (C 1s) at 284.6 eV. The XPS analysis was performed at ambient temperature and at pressures typically on the order of $< 10^{-8}$ Torr. Prior to the analysis, the samples were out gassed under vacuum for 4 h.

2.3. Catalytic activity

Syngas conversion was performed in a stainless steel in a vertical down flow fixed-bed reactor at a total pressure of 350 psig with a H₂/CO molar ratio of 2/1. Typically, the required amount of catalyst placed in the center of the reactor, and the catalyst bed was positioned between plugs of quartz wool. The reactor was filled with silica beads to the top and bottom of the heating zone. The reactor was placed vertically inside a programmable tubular furnace (Lindberg). The reactor was heated externally via a tubular furnace regulated by a temperature controller (Omega CN 2041), with a thermocouple inserted into the catalyst bed. The gas flow was controlled by a Brooks mass flow meter. The pressure of the reactor was maintained by the mass flow controllers and back pressure regulators. Prior to the reaction, the catalyst was *in-situ* reduced in hydrogen with 50 ml/min flow rate at the atmospheric pressure at 400 °C for 1 h. After reduction, the reactor was cooled to the desired reaction temperature and switched to a certified H₂/CO mixture. Then, the system was gradually pressurized to 350 psig with the reactants gas flow. Then the reactants stream was passed with space velocity (S.V.) of 6,000 sccm g⁻¹h⁻¹. The products of the reaction were analyzed on-line by using HP gas chromatograph.

The selectivity for products is reported as atomic carbon selectivity (Equation (1)), which accounts for the number of carbon atoms inherent in the molecule as well as the concentration.

$$\text{Carbon efficiency} = n_i C_i / \sum (n_i C_i) \quad (1)$$

n_i = no. of carbon atoms presented in that species i
 C_i = no. of moles of species i in the product steam

3. Results and discussion

The CO conversion and selectivity for various products in the syngas to oxygenates reaction over various co-promoted Rh/SiO₂ catalysts as a function of reaction temperature in the range of 260–320 °C are shown in Fig. 1. The detailed activity and selectivity results are also tabulated in Tables 1–4. As can be observed, the reaction temperature has a significant effect on the activity and selectivity of the catalysts. The CO conversion of all catalysts didn't alter much in the temperature range of 260–280 °C. After 280 °C, the activity of RLWS moderately increased, while the RLVS catalyst exhibited dramatic enhancement in the CO conversion. Moreover, the CO conversion of RLWS and RLVS catalysts reached to about 98% and about 94%, respectively, at 320 °C. On the other hand, the activity of RLCS and RLZS catalysts slightly increased when the reaction temperature increased to 300 °C. With further increasing the reaction temperature to 320 °C, the CO conversion of RLCS reached to about 68%, while the RLZS showed about 98% of CO conversion. It is clear that the RLWS and RLVS catalysts displayed

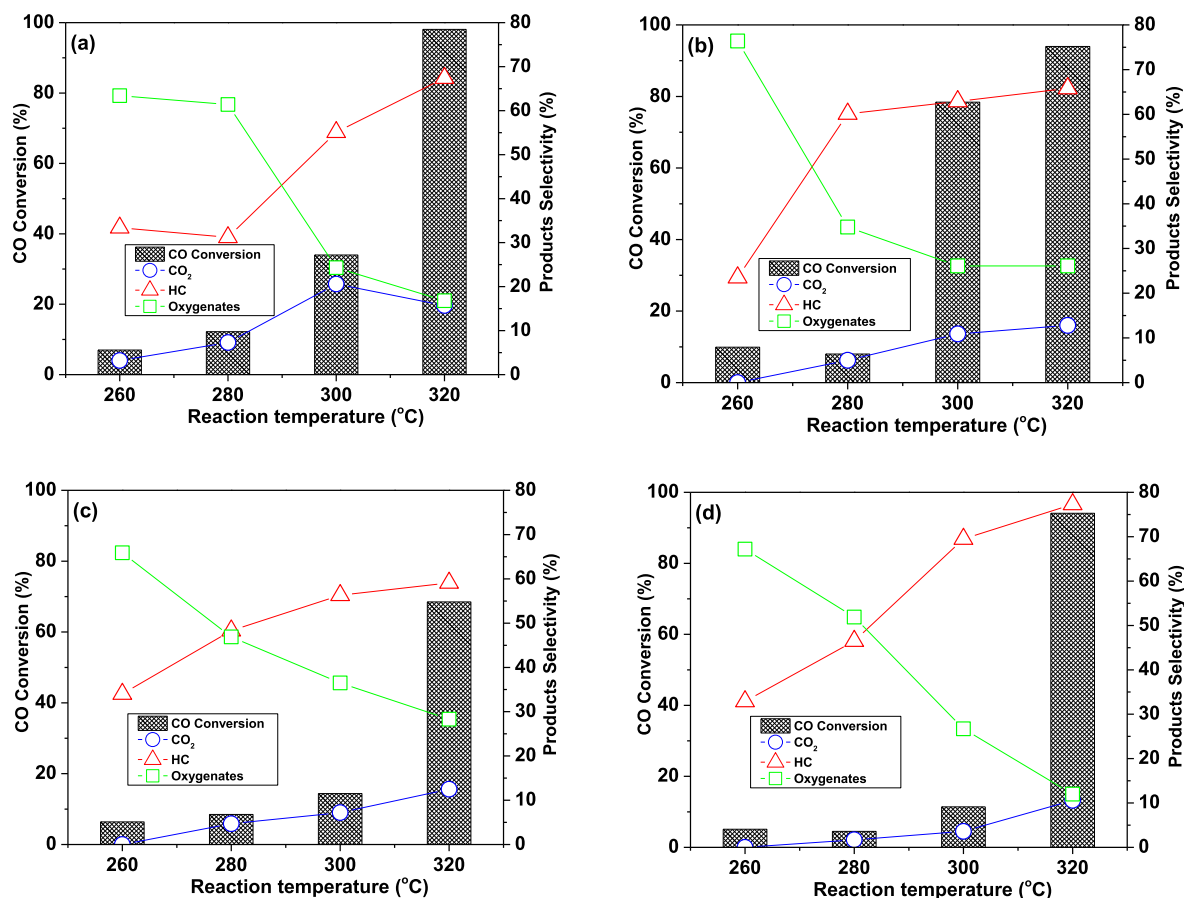


Fig. 1. CO conversion and products selectivity over (a) Rh-La(0.8)/1.8 W/SiO₂, (b) Rh-La(0.8)/1.8 V/SiO₂, (c) Rh-La(0.8)/1.8Ce/SiO₂, and Rh-La(0.8)/1.8Zr/SiO₂ catalysts.

Table 1

CO conversion and products selectivity over Rh-La(0.8)/1.8W/SiO₂^a catalyst.

T (°C)	X _{CO} %	Selectivity (% C) ^b						
		CH ₄	CO ₂	MeOH	EtOH	Oxy ^c	HC ^d	C ₂₊ Oxy ^e
260	7.0	23.7	3.2	6.3	20.8	63.4	33.4	12.5
280	12.2	25.5	7.3	5.3	34.2	61.4	31.2	0.0
300	34.0	46.9	20.6	6.1	4.2	24.3	55.1	1.1
320	98.1	65.0	15.7	2.3	2.3	16.8	67.4	0.2

Reaction Conditions: P = 350 psi, H₂/CO = 2, S.V. = 6,000 mLgcat⁻¹h⁻¹.

^a The numbers in the parenthesis indicate the Rh/La atomic ratio and nominal weight percentage of third metal relative to the initial weight of the support material.

^b Product selectivities are reported in terms of carbon efficiencies defined as: carbon efficiency = $n_i C_i / \sum (n_i C_i)$, where n_i is the number of carbon atoms and C_i is the concentration of the carbon-containing products.

^c Oxygenates including MeOH, EtOH, and C₂₊ oxygenates, except carbon dioxide (CO₂).

^d Hydrocarbons including methane.

^e Oxygenates with two or more carbons except ethanol (acetaldehyde, i-propanol, i-butanol, n-propanol, and n-butanol).

Table 2

CO conversion and products selectivity over Rh-La(0.8)/1.8V/SiO₂^a catalyst.

T (°C)	X _{CO} %	Selectivity (% C) ^b						
		CH ₄	CO ₂	MeOH	EtOH	Oxy ^c	HC ^d	C ₂₊ Oxy ^e
260	9.9	20.7	0.0	8.1	50.1	76.5	23.5	5.7
280	8.0	57.1	5.0	15.9	14.0	34.8	60.1	3.6
300	78.4	58.6	10.9	10.9	5.1	26.1	62.9	1.5
320	94.0	61.9	12.8	7.3	3.1	21.3	65.8	1.2

Reaction Conditions: P = 350 psi, H₂/CO = 2, S.V. = 6,000 mLgcat⁻¹h⁻¹.

^a The numbers in the parenthesis indicate the Rh/La atomic ratio and nominal weight percentage of third metal relative to the initial weight of the support material.

^b Product selectivities are reported in terms of carbon efficiencies defined as: carbon efficiency = $n_i C_i / \sum (n_i C_i)$, where n_i is the number of carbon atoms and C_i is the concentration of the carbon-containing products.

^c Oxygenates including MeOH, EtOH, and C₂₊ oxygenates, except carbon dioxide (CO₂).

^d Hydrocarbons including methane.

^e Oxygenates with two or more carbons except ethanol (acetaldehyde, i-propanol, i-butanol, n-propanol, and n-butanol).

Table 3CO conversion and products selectivity over Rh-La(0.8)/1.8Ce/SiO₂^a catalyst.

T (°C)	X _{CO} %	Selectivity (% C) ^b						
		CH ₄	CO ₂	MeOH	EtOH	Oxy ^c	HC ^d	C ₂₊ Oxy ^e
260	6.4	26.7	0.0	11.6	20.9	65.9	34.0	8.6
280	8.5	38.6	4.7	10.8	15.8	46.9	48.3	4.8
300	14.4	46.1	7.2	14.4	9.2	36.5	56.3	7.3
320	68.5	52.4	12.5	11.7	4.3	28.3	59.1	1.8

Reaction Conditions: P = 350 psi, H₂/CO = 2, S.V. = 6,000 mLgc⁻¹h⁻¹.^a The numbers in the parenthesis indicate the Rh/La atomic ratio and nominal weight percentage of third metal relative to the initial weight of the support material.^b Product selectivities are reported in terms of carbon efficiencies defined as: carbon efficiency = $n_i C_i / \sum (n_i C_i)$, where n_i is the number of carbon atoms and C_i is the concentration of the carbon-containing products.^c Oxygenates including MeOH, EtOH, and C₂₊ oxygenates, except carbon dioxide (CO₂).^d Hydrocarbons including methane.^e Oxygenates with two or more carbons except ethanol (acetaldehyde, i-propanol, i-butanol, n-propanol, and n-butanol).**Table 4**CO conversion and products selectivity over Rh-La(0.8)/1.8Zr/SiO₂^a catalyst.

T (°C)	X _{CO} %	Selectivity (% C) ^b						
		CH ₄	CO ₂	MeOH	EtOH	Oxy ^c	HC ^d	C ₂₊ Oxy ^e
260	5.1	28.6	0.0	9.0	22.8	67.1	32.9	4.6
280	4.5	46.5	1.7	12.4	22.1	51.8	46.5	9.2
300	11.4	62.1	3.6	12.5	6.9	26.7	69.5	2.9
320	94.1	74.2	10.6	2.7	1.0	12.0	77.3	0.2

Reaction Conditions: P = 350 psi, H₂/CO = 2, S.V. = 6,000 mLgc⁻¹h⁻¹.^a The numbers in the parenthesis indicate the Rh/La atomic ratio and nominal weight percentage of third metal relative to the initial weight of the support material.^b Product selectivities are reported in terms of carbon efficiencies defined as: carbon efficiency = $n_i C_i / \sum (n_i C_i)$, where n_i is the number of carbon atoms and C_i is the concentration of the carbon-containing products.^c Oxygenates including MeOH, EtOH, and C₂₊ oxygenates, except carbon dioxide (CO₂).^d Hydrocarbons including methane.^e Oxygenates with two or more carbons except ethanol (acetaldehyde, i-propanol, i-butanol, n-propanol, and n-butanol).

better activity than the RLCS and RLZS at temperature below 300 °C.

The incorporation of different promoters to Rh/SiO₂ also led to a significant change in the product selectivity. At 260 °C, the RLVS showed the highest oxygenates selectivity (76.5%) by significantly suppressing the hydrocarbon formation (23.5%) among the investigated catalysts. However, above 260 °C, the oxygenates selectivity of RLVS has dramatically reduced as increasing the reaction temperature. At the same time, this catalyst exhibited a large enhancement in hydrocarbons production. This indicates that the decrease in selectivity for oxygenates over the RLVS catalyst is accompanied by an increase in hydrocarbon formation. RLWS showed about 63.4% oxygenates selectivity at 260 °C and a slight decrease in the selectivity towards oxygenates is observed when the reaction temperature increased to 280 °C. This decline in the oxygenates selectivity could be due to the increasing CO₂ selectivity, as the selectivity to hydrocarbons slightly reduced from 260 °C to 280 °C. With further increasing the reaction temperature, the selectivity of hydrocarbons has rapidly enhanced over the RLWS at the expense of oxygenates selectivity. The RLCS and RLZS catalysts also resulted in a good amount of oxygenates (65.9% and 67.2%, respectively) with a relatively low fraction of hydrocarbons (34.0% and 32.9%, respectively) at 260 °C. In addition, the selectivity of hydrocarbons moderately enhanced from 260 °C to 280 °C, which significantly augmented with further increasing the reaction temperature. Meanwhile, the oxygenates selectivity of RLCS and RLZS also dropped which is accompanied by an increase in the fraction of hydrocarbons. Based on the above findings, it can be concluded that the oxygenates formation favorable at lower reaction temperatures (≤ 280 °C) by suppressing hydrocarbons on all the catalysts investigated. The oxygenates selectivity for all the catalysts decreased with increasing reaction temperature, however, the selectivity over RLWS, RLCS, and RLZS changed little to moder-

ate at below 300 °C. Typically, the oxygenates formation facilitated by both CO dissociation and CO insertion steps (Chuang et al., 2005). It has been reported based on isotopic tracer studies that the CO dissociation does not occur readily over un-promoted Rh catalyst, even at elevated temperatures (Winslow and Bell, 1985). Hence, the higher oxygenates selectivity of catalysts possibly resulted from the promotional effect of La and W/V/Ce/Zr metals in enhancing the CO dissociation and insertion (Poniec, 1991).

At 260 °C, no CO₂ is produced from the RLVS, RLCS, and RLZS catalysts, while the CO₂ formation is visible over the RLWS catalyst. However, all the catalysts have generated CO₂ above 260 °C. The CO₂ formation possibly resulted from the water–gas shift reaction, which is in agreement with literature reports (Haider et al., 2009; Yao et al., 2017). In addition, the CO₂ selectivity of all catalysts increased as increasing the reaction temperature. This could be due to the enhanced water–gas shift reaction by the increased content of water that comes from the further reaction of oxygenates at higher temperatures. Nevertheless, it should be noted that the CO₂ is not a major unwanted by-product, as it can be easily recycled to regenerate CO through reverse water–gas shift. Therefore, the production of hydrocarbon is regarded to be the main undesirable side product in the syngas to oxygenates reaction.

Further, the effect of space velocity is investigated for the RLVS catalyst in the range of 2,000–10,000 sccm gc⁻¹h⁻¹ at 260 °C and 350 psi total pressure with H₂/CO ratio of 2. As can be seen from Fig. 2 and Table 5, the CO conversion decreases from 12.3% to 3.2% with increasing the space velocity from 2,000 to 10,000 sccm gc⁻¹h⁻¹. This can be understood from the lower residence time of reactants over the surface of the catalyst at higher space velocities. The selectivity toward oxygenates increased monotonically up to 6,000 sccm gc⁻¹h⁻¹ and then decreased with further increase in space velocity. The greater oxygenates selectivity at 6,000 sccm gc⁻¹h⁻¹ is apparently due to a higher relative rate of CO

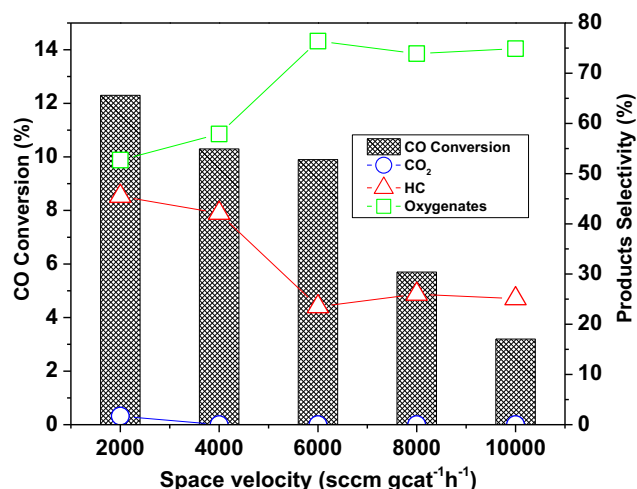


Fig. 2. Effect of space velocity on the CO conversion and products selectivity of Rh-La(0.8)/1.8 V/SiO₂ catalyst at 260 °C.

dissociation and/or insertion. Conversely, hydrocarbon selectivity gradually decreased up to 6,000 sccm gcat⁻¹h⁻¹ and then increased as increasing the space velocity. It is obvious that the selectivities of oxygenates and hydrocarbons followed the opposite trend with space velocity, indicating that they are produced by parallel reactions, which is in harmony with previous reports (Spivey and Egbebi, 2007). This also suggests that the hydrocarbons selectivity increases at the expense of oxygenates at higher space velocities (>6,000 sccm gcat⁻¹h⁻¹). Moreover, the selectivity of CO₂ dropped to zero after the space velocity of 2,000 sccm gcat⁻¹h⁻¹. This

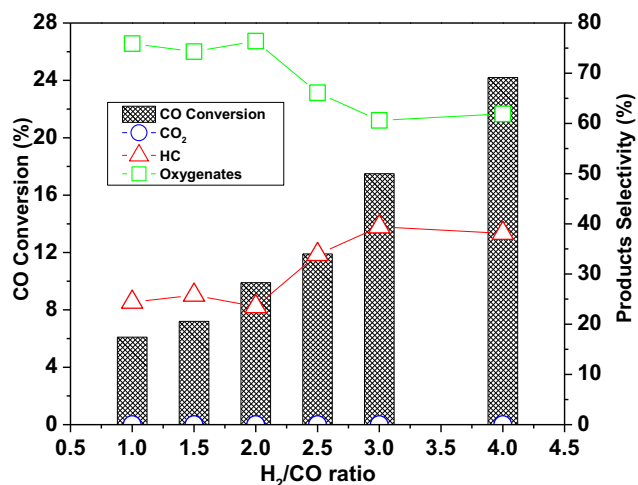


Fig. 3. Effect of H₂/CO ratio on the CO conversion and products selectivity of Rh-La(0.8)/1.8 V/SiO₂ catalyst at 260 °C.

implies that the higher space velocities didn't favor the water-gas shift reaction during the syngas conversion.

The effect of the H₂/CO ratio on the activity and selectivity of RLVS is also investigated at 260 °C with a space velocity of 6,000 sccm gcat⁻¹h⁻¹ and corresponding results are illustrated in Fig. 3 and Table 6. The CO conversion gradually increased when the H₂/CO ratio was increased from 1 to 4. No CO₂ formation was observed at all the H₂/CO ratios. This suggests that the water-gas shift reaction was not instigated over the RLVS catalyst under the tested

Table 5

Effect of space velocity on the CO conversion and products selectivity of Rh-La(0.8)/1.8 V/SiO₂^a catalyst at 260 °C.

Space velocity (mLgcat ⁻¹ h ⁻¹)	X _{CO} %	Selectivity (% C) ^b						
		CH ₄	CO ₂	MeOH	EtOH	Oxy ^c	HC ^d	C ₂₊ Oxy ^e
2,000	12.3	41.0	1.7	10.4	13.9	52.7	45.5	3.7
4,000	10.3	38.3	0.0	9.3	21.2	57.9	42.1	5.4
6,000	9.9	20.7	0.0	8.1	50.1	76.5	23.5	5.7
8,000	5.7	22.5	0.0	10.0	40.2	73.9	26.0	8.5
10,000	3.2	22.2	0.0	8.8	37.3	74.9	25.1	11.6

Reaction Conditions: P = 350 psi, H₂/CO = 2.

^a The numbers in the parenthesis indicate the Rh/La atomic ratio and nominal weight percentage of third metal relative to the initial weight of the support material.

^b Product selectivities are reported in terms of carbon efficiencies defined as: carbon efficiency = $n_i C_i / \sum (n_i C_i)$, where n_i is the number of carbon atoms and C_i is the concentration of the carbon-containing products.

^c Oxygenates including MeOH, EtOH, and C₂₊ oxygenates, except carbon dioxide (CO₂).

^d Hydrocarbons including methane.

^e Oxygenates with two or more carbons except ethanol (acetaldehyde, i-propanol, i-butanol, n-propanol, and n-butanol).

Table 6

Effect of H₂/CO ratio on the CO conversion and products selectivity of Rh-La(0.8)/1.8 V/SiO₂^a catalyst at 260 °C.

H ₂ /CO ratio	X _{CO} %	Selectivity (% C) ^b						
		CH ₄	CO ₂	MeOH	EtOH	Oxy ^c	HC ^d	C ₂₊ Oxy ^e
1.0	6.1	21.9	0.0	7.4	28.9	75.6	24.4	13.1
1.5	7.2	22.8	0.0	7.7	28.6	74.2	25.8	9.0
2.0	9.9	20.8	0.0	8.2	50.1	76.5	23.5	5.8
2.5	11.9	30.9	0.0	6.3	28.2	66.1	33.9	8.1
3.0	17.5	36.4	0.0	6.1	26.0	60.6	39.4	5.2
4.0	24.2	35.8	0.0	4.1	18.5	61.9	38.1	9.9

Reaction Conditions: P = 350 psi, S.V. = 6,000 mLgcat⁻¹h⁻¹.

^a The numbers in the parenthesis indicate the Rh/La atomic ratio and nominal weight percentage of third metal relative to the initial weight of the support material.

^b Product selectivities are reported in terms of carbon efficiencies defined as: carbon efficiency = $n_i C_i / \sum (n_i C_i)$, where n_i is the number of carbon atoms and C_i is the concentration of the carbon-containing products.

^c Oxygenates including MeOH, EtOH, and C₂₊ oxygenates, except carbon dioxide (CO₂).

^d Hydrocarbons including methane.

^e Oxygenates with two or more carbons except ethanol (acetaldehyde, i-propanol, i-butanol, n-propanol, and n-butanol).

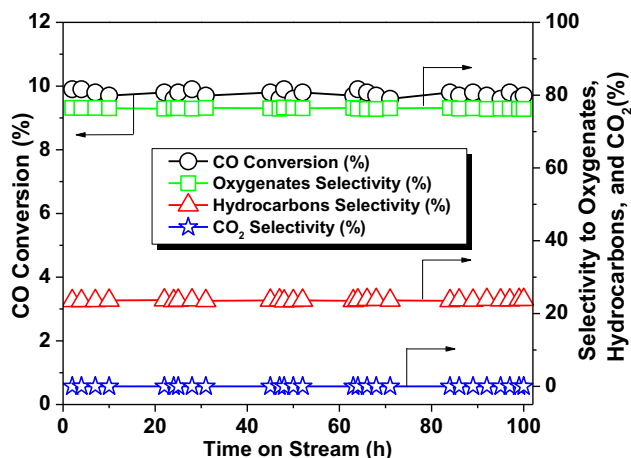


Fig. 4. CO conversion and products selectivity for Rh-La(0.8)/1.8 V/SiO₂ catalyst with time on stream (Reaction conditions: T = 260 °C, P = 350 psi, S.V. = 6,000 mL_{g_{cat}}⁻¹h⁻¹).

conditions. Interestingly, no significant change was observed in the selectivity of either oxygenates or hydrocarbons as increasing the H₂/CO ratio to 2 from 1. A further increase in the H₂/CO ratio led to an increase in the selectivity of hydrocarbons at the expense of oxygenates selectivity. This suggests that the increase in CO conversion is accompanied by an increase in the selectivity to hydrocarbons when the H₂/CO ratio was raised from 2 to 4. From these findings, it can be concluded that the RLVS catalyst suppresses the hydrocarbons formation appreciably at lower H₂/CO ratios.

Besides, a long-term test was also performed for the RLVS catalyst to verify its stability with time on stream. As can be observed from Fig. 4, the RLVS catalyst displayed a stable performance in terms of both CO conversion and product selectivity during 100 h of the reaction. In the reaction period of 100 h, the conversion of CO and the selectivity of oxygenates were sustained at 9.7% and 76.2%, respectively. This finding suggested that the RLVS catalyst exhibited an excellent time on stream stability for the synthesis of oxygenates from syngas. The catalytic performance of the RLVS catalyst was also compared with other Rh-based promoted catalysts reported in the literature. The CO conversion and selectivity for various products as well as the operating conditions used in testing of different catalyst systems have been illustrated in Table 7. As evident from Table 7, the present RLVS catalyst has a comparable activity with that of catalysts reported in the literature. Most importantly, it is shown better performance in terms of oxygenates selectivity as compared to the similar catalysts found in the literature. Hence, the RLVS catalyst seems to be very promising to produce oxygenates from syngas.

The textural properties of co-promoted Rh/SiO₂ catalysts were determined by N₂ sorption method. As shown in Fig. 5, all the sam-

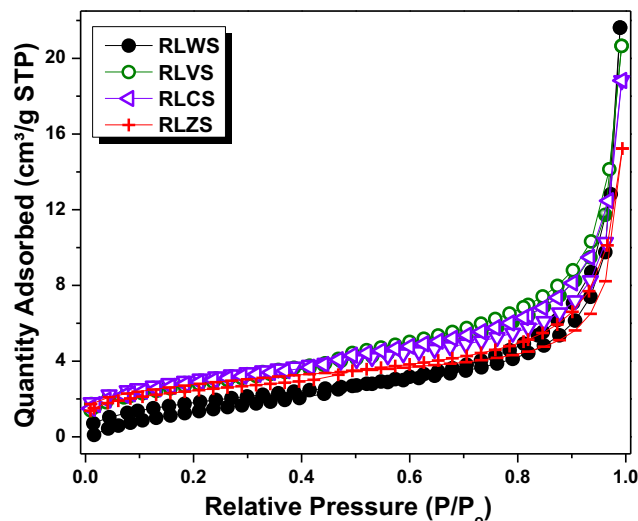


Fig. 5. Nitrogen adsorption-desorption isotherms of fresh Rh-La(0.8)/1.8 W/SiO₂, Rh-La(0.8)/1.8 Ce/SiO₂, and Rh-La(0.8)/1.8 Zr/SiO₂ catalysts.

ples showed typical type-IV adsorption-desorption isotherms with a small hysteresis loop, indicating the existence of porous structure in the catalysts. Moreover, isotherms of all catalysts displayed the N₂ capillary condensation in a wide relative pressure of 0.45–0.99, which represent the mesoporous structures (Devaiah et al., 2015). This mesoporosity may have originated from the inter connections between agglomerates existed in the catalysts. The surface area, pore diameter, pore volume for the catalysts is in the range of 7.0–10.5 m²/g, 10.1–18.9 nm, and 0.023–0.033 cm³/g, respectively (Table 8). This implies that there is no significant difference in the textural properties possibly due to the fact that the concentrations of Rh and promoters were relatively low in all the catalysts prepared (Gao et al., 2009). Moreover, it is apparent that all catalysts exhibited low surface area, which is consistent with previous

Table 8
BET surface area, pore diameter, and pore volume of the catalysts.

Catalyst	S _{BET} (m ² /g)	Average pore diameter (nm)	Isotherm type	Pore volume (cm ³ /g)
RLWS ^a	7.0	18.9	IV	0.032
RLVS ^b	10.3	12.4	IV	0.033
RLCS ^c	10.5	11.0	IV	0.029
RLZS ^d	9.3	10.1	IV	0.023

^a RLWS: Rh-La(0.8)/1.8 W/SiO₂; ^bRLVS: Rh-La(0.8)/1.8 V/SiO₂; ^cRLCS: Rh-La(0.8)/1.8 Ce/SiO₂; ^dRLZS: Rh-La(0.8)/1.8 Zr/SiO₂; the numbers in the parenthesis indicate the Rh/La atomic ratio and nominal weight percentage of third metal relative to the initial weight of the support material, respectively.

Table 7
Comparison of Rh-La(0.8)/1.8 V/SiO₂ catalyst with the similar Rh-based catalysts found in the literature.

Catalyst	Temperature & Pressure	Space velocity & H ₂ /CO ratio	X _{CO} %	Selectivity (% C)			Ref.
				CO ₂	HC ^a	Oxy ^b	
2 wt.%Ce-4 wt.%Fe-2wt.%Rh/TiO ₂	260 °C & 2 Mpa	3600 ml(g·h) ⁻¹ & H ₂ /CO = 2	12.6	12.3	51.4	36.3	Xu et al. (2017)
3 wt.%Ce-0.5 wt.%Fe-2 wt.% Rh/SiO ₂	290 °C & 3 Mpa	5000 ml(g·h) ⁻¹ & H ₂ /CO = 2	6.2	5.5	34.2	56.7	Liu et al. (2016)
30 wt.%La _{0.9} Ce _{0.1} Fe _{0.5} Rh _{0.5} O ₃ /SiO ₂ (where Rh loading amount was fixed at 1.0 wt.%)	250 °C & 3 Mpa	3900 ml(g·h) ⁻¹ & H ₂ /CO = 2	14.2	2.0	29.4	68.6	Zhong et al. (2021)
1 wt.%Rh-0.2 wt.%Fe/SiO ₂	280 °C & 5 Mpa	5000 h ⁻¹ & H ₂ /CO = 2	5.4	0.8	44.2	55.0	Xue et al. (2016)
2 wt.%Rh-2 wt.%Mn/TiO ₂ -anatase rich	280 °C & 3 Mpa	10000 h ⁻¹ & H ₂ /CO = ~2	6.8	13.0	45.1	41.9	Yu et al.
1.5 wt.%Rh-2.5 wt.%La/1.8 wt.%V/SiO ₂	260 °C & 2.4 Mpa	6000 h ⁻¹ & H ₂ /CO = 2	9.9	0.0	23.5	76.5	This work

^a Oxygenates including MeOH, EtOH, and C₂₊ oxygenates, except carbon dioxide (CO₂).

^b Hydrocarbons including methane.

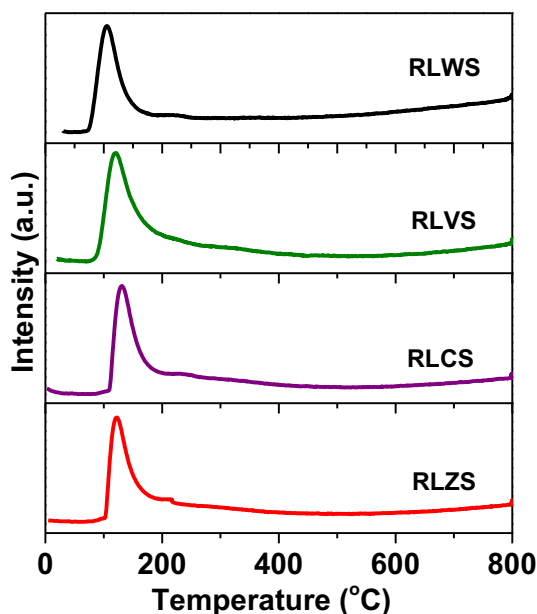


Fig. 6. H_2 -TPR profiles of fresh (a) Rh-La(0.8)/1.8 W/SiO₂, (b) Rh-La(0.8)/1.8 V/SiO₂, (c) Rh-La(0.8)/1.8Ce/SiO₂, and Rh-La(0.8)/1.8Zr/SiO₂ catalysts.

reports on Rh/SiO₂ based catalysts (Yu et al., 2016; Ding et al., 2017). This demonstrates that the surface area of the catalysts seems that it does not benefit their catalytic performance.

The reducibility of the catalysts was investigated by hydrogen temperature programmed reduction (H_2 -TPR) experiments. The H_2 -TPR profiles of the co-promoted Rh/SiO₂ catalysts are illustrated in Fig. 6. As can be observed, all the samples exhibited a sharp reduction peak in the temperature range of 106–130 °C. This peak corresponds to the reduction of rhodium oxide to its metallic phase (Sun et al., 2020). The sharp reduction peak indicates the narrow size distribution of Rh particles in the catalysts. Typically, the reduction of Rh species has been reported at below 100 °C for un-promoted Rh/SiO₂ catalyst (Mo et al., 2009; Yang et al., 2017; Lopez et al., 2015). It implies that the addition of promoters to the Rh/SiO₂ led to a shift of the Rh reduction to higher temperatures above 100 °C, which is consistent with literature reports (Mo et al., 2009; Yang et al., 2017). This further indicates the strong interaction between Rh and promoters in the catalysts which could contribute to the enhanced activity and selectivity towards oxygenates. Another weak broad reduction peak evolution can be observed in the temperature region of 200–450 °C for all co-promoted Rh/SiO₂ catalysts, which can be attributed to the existence of LaRhO₃. This observation is in good agreement with that of Tascon et al. (Tascon et al., 1986).

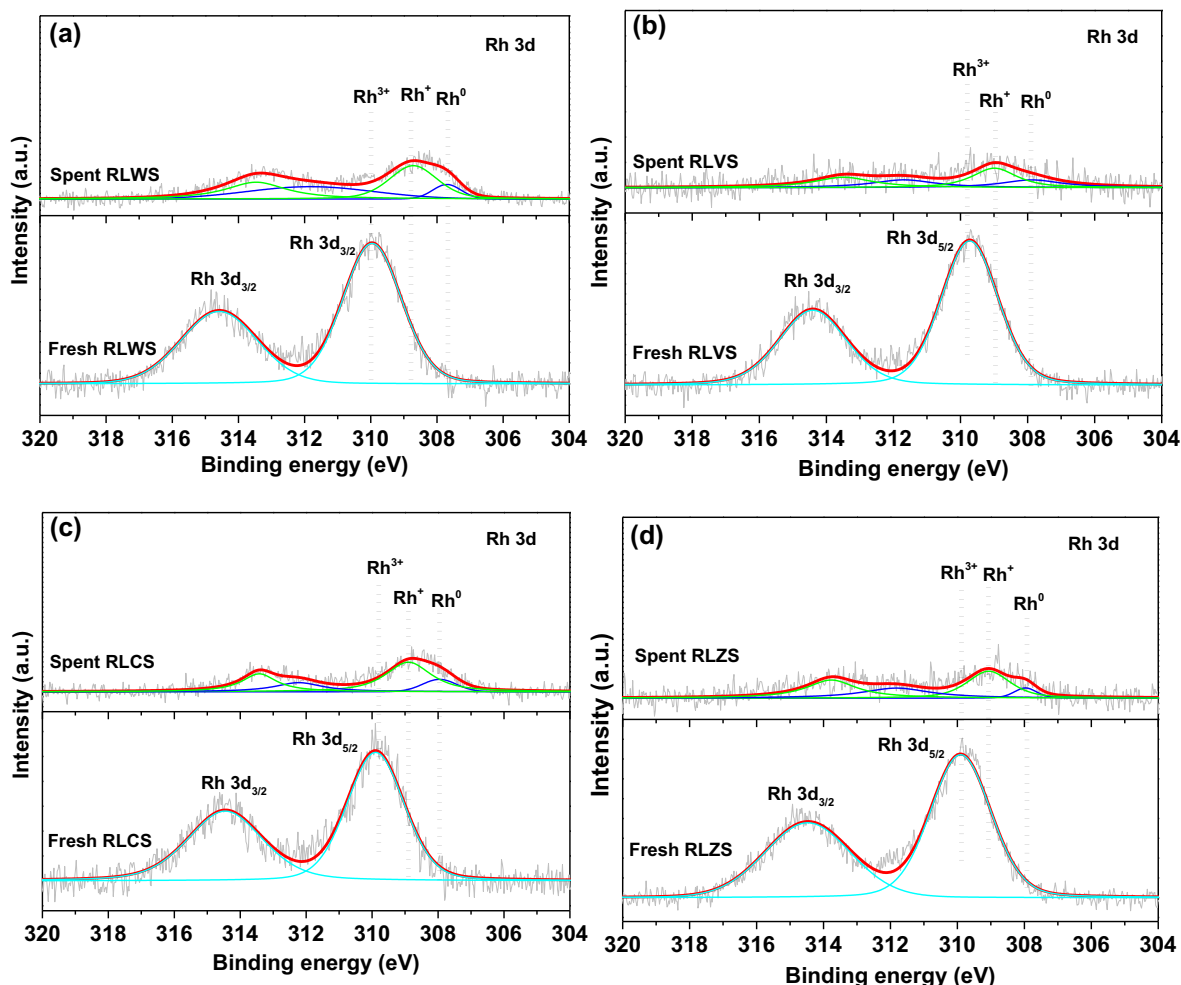


Fig. 7. Rh 3d XPS spectra of fresh and spent (a) Rh-La(0.8)/1.8 W/SiO₂, (b) Rh-La(0.8)/1.8 V/SiO₂, (c) Rh-La(0.8)/1.8Ce/SiO₂, and Rh-La(0.8)/1.8Zr/SiO₂ catalysts.

The surface chemical state of Rh on the surface of the catalysts was investigated by XPS study. The Rh 3d XPS spectra of fresh and spent catalysts are presented in Fig. 7. The Rh spectrum of all fresh samples is composed of two distinct peaks at 309.7–309.9 eV and 314.4–314.7 eV which are attributed to Rh 3d_{5/2} and Rh 3d_{3/2} states, respectively (Abdelsayed et al., 2013). The single Rh 3d_{5/2} peak at 309.7–309.9 eV indicates that the Rh existed as Rh₂O₃ in all the fresh samples (Ding et al., 2017; Yu et al., 2019). It can be seen that the Rh 3d peaks of spent catalysts shifted to lower binding energies as compared to the fresh samples. This suggests that the electron density on Rh increased which could be due to the reduction of Rh₂O₃ species on the surface of catalysts during the reaction. To verify this observation, the Rh 3d_{5/2} sublevel of all the spent catalysts was deconvoluted. After deconvolution, Rh 3d_{5/2} spectra split into two components at 307.6–307.9 eV and 308.7–309.0 eV, which corresponded to the Rh⁰ and Rh⁺ species, respectively (Han et al., 2013). This confirms that the Rh oxide got reduced to both the Rh⁺ and Rh⁰ states under the reaction conditions. Moreover, the intensity of the Rh⁺ peak is higher than the Rh⁰ peak, indicating that the Rh⁺ is the major Rh species on the surface of the spent catalysts. The dominant Rh⁺ species seems to be an important factor for the higher oxygenates productivity of catalysts (Ojeda et al., 2004; Won et al., 2016). The existence of oxidized Rh (Rh⁺) as major species on the surface of spent catalysts could be indicative of a strong synergistic interaction between Rh and promoters that effectively retarded the complete reduction of Rh₂O₃ to metallic Rh species during the reaction. This is in well agreement with our H₂-TPR results.

4. Conclusions

A series of Rh-La/M/SiO₂ (RLMS, where M = W, V, Ce, and Zr) catalysts were prepared by using a co-sequential incipient wetness impregnation method and tested for the syngas to oxygenates reaction. No appreciable change in the CO conversion of catalysts was observed from 260 °C to 280 °C. However, after 280 °C, the catalysts, especially RLVS, exhibited significant enhancement in the CO conversion with reaction temperature. The selectivity towards oxygenates formation over all the catalysts was superior at the lower reaction temperatures (≤280 °C). Among the investigated catalysts, the RLVS exhibited the greatest selectivity towards oxygenates (76.5%) with CO conversion of 9.9% at 260 °C. After 260 °C, this catalyst experienced a significant loss in the oxygenates selectivity with reaction temperature. At 280 °C, RLWS, RLCS, and RLZS samples owned higher oxygenates selectivity than that of RLVS catalyst. The rhodium oxide, which existed on the surface of the fresh catalysts, was found to be reduced to Rh⁺ and Rh⁰ species during the reaction. The strong synergistic interaction between Rh and promoters could be responsible for the better oxygenates selectivity of the catalysts.

CRedit authorship contribution statement

Devaiah Damma: Conceptualization, Data curation, Formal analysis, Investigation, Validation, Writing - original draft, Writing - review & editing. **Thirupathi Boningari:** Conceptualization, Data curation, Formal analysis, Investigation, Validation. **Panagiotis G. Smirniotis:** Supervision, Validation, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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